

Total No. of Questions : 4]

SEAT No. :

PF304

[Total No. of Pages : 1

Apr-26/TE/Insem-384

T.E. (Artificial Intelligence and Data Science) (Insem)

DATA SCIENCE

(2019 Pattern) (Semester - VI) (317529)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Solve Q1 or Q2 and Q3 or Q4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) What is Data Science? Differentiate between Data Science and Information Science? [5]
b) List and explain data reduction techniques with example. [5]
c) Explain Data Integration and Discretization. [5]

OR

- Q2)** a) What is data wrangling? Explain various methods of data wrangling. [5]
b) What is Data? Explain different types of Data and Data Collection. [5]
c) Explain the various applications of Data Science. [5]

- Q3)** a) What is Hypothesis? Explain null and alternative hypothesis with an example. [5]
b) Describe Bayes theorem in detail with an example. [5]
c) In a garden, there are 39 plants. The heights (in cm) of 5 randomly selected plants are 38, 51, 46, 79, and 57. Calculate the standard deviation of their heights. [5]

OR

- Q4)** a) Explain need of statistics in data science. [5]
b) Explain different measures of dispersion. [5]
c) Solve the following. [5]

The average score of a class is 90. However, a teacher believes that the average score might be lower. The scores of 6 students were randomly measured. The mean was 82 with a standard deviation of 18. With a 0.05 significance level use hypothesis testing to check if this claim is true.

x

x

x